

Level and Shape: Learning Adjacent-Lanthanide Selectivity from the Solvent-Extraction Record

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September 12, 2026

Abstract

Adjacent lanthanide ions differ in radius by about 0.013 Å, and separating them is the most expensive step in rare-earth processing. In this work, we show that the published solvent-extraction record contains sufficient information to learn which ligands perform this separation effectively. On 4,746 measurements covering 162 extractants, a gradient-boosting model evaluated with whole extractants held out predicts log separation factors that correlate with experiment at $r = 0.59$ across 905 adjacent pairs. The prediction is split in two: one model estimates the overall extraction level of each series, the other the differences between its metals, and the pair outperforms a single model trained directly on the separation factors. Used as a screen, the model identifies the most effective separations (above a factor of two) with an AUC of 0.81, and extractants ranked in the model’s top quartile fall within the true top quartile 50 % of the time, twice the rate expected by chance. The conformal prediction intervals match their nominal coverage. A graph neural network operating on three-dimensional metal–ligand geometries provides similar predictions independently complex and adds a small further gain in combined model.

1 Introduction

Trivalent lanthanides scarcely differ chemically. Along the series, the ionic radius decreases by about 0.013 Å from one lanthanide to the next [1]. A ligand that separates neodymium from praseodymium has, in effect, mostly only that shrinkage to use for the separation. An industrial plant therefore runs its extraction stages in long series. New extractants still found, for the most part, from trial and error [2].

The SAFE database [3] compiles distribution ratios from the solvent-extraction literature, and many of its ligands have been measured across most of the series under a single set of conditions. The separation factor of a pair of metals is the ratio of their distribution ratios from the same experiment, so errors that are common to both measurements cancel. This paper addresses how much adjacent-lanthanide selectivity can be learned from this record for extractants not seen during training, and what a three-dimensional (3D) structure of the metal–ligand complex adds. The work makes the following three contributions: (i) an evaluation score and a model that both separate extraction strength from selectivity ($+0.27 \rightarrow +0.33$ from the model split with the same learner and features); (ii) a small gain from 3D structures of the complexes;

and (iii) a benchmark of three tight-binding Hamiltonians on the lanthanide contraction, which measures how much information about the metal a computed geometry contains.

2 Data and target

From the SAFE Database lanthanide(III) rows with a resolvable extractant structure were kept. We canonicalise the SMILES, remove duplicates and detection-limit placeholders, and retain the 4,746 measurements of $\log D$ on 162 extractants for which a complex geometry exists. The 956 complexes were built with Architector [4] (extractant and nitrate or water to coordination number 8–10) and optimised with GFN2-xTB [5].

A separation factor is only defined between measurements made under the same conditions, therefore rows are grouped into blocks (one extractant, one bin of each of ten condition descriptors). Within a block, replicate $\log D$ measurements of one metal are averaged into a cell, and each pair of cells whose lanthanides lie adjacent in the series contributes one target value,

$$\log \text{SF}_{i,i+1} = \log D_{\text{Ln}_i} - \log D_{\text{Ln}_{i+1}}. \quad (1)$$

Models are trained on $\log D$ row by row and evaluated after this construction is applied to their out-of-fold predictions; the score R_{adj}^2 is the coefficient of determination of predicted against measured $\log \text{SF}$ over all adjacent pairs.

Every number presented below is out-of-fold under five-fold cross-validation grouped by extractant, repeated three times. The population yields 905 adjacent pairs on 76 extractants.

3 Model

The tabular features are RDKit descriptors and 2048-bit Morgan fingerprints of the extractant [6], the metal’s atomic number, position in the series and Shannon radius, and the 64 condition columns, 746 in total. The learner is CatBoost [7] with a quantile loss ($\alpha = 0.6$), depth 9 and feature subsampling 0.3, chosen on the adjacent-pair score.

The raw target combines the overall extraction strength of a ligand with its variation across metals. Only the variation is scored, so each part receives its own model. The level model f_L is trained on $\log D$, and only its test-block mean, the anchor a_b , is used. The shape model f_S is trained on the block-centred target $\log D_i - \bar{y}_{b(i)}$, and its predictions are centred within the test block in the same way,

$$\hat{y}_i = a_{b(i)} + \left(f_S(x_i) - \text{mean}_{j \in b(i)} f_S(x_j) \right). \quad (2)$$

For the 3D channel, heavy atoms of the complex are nodes, atoms within 4 Å are connected, and three continuous-filter layers of width 96 [8] feed a readout with a metal-centred term and a radial shell histogram, since plain mean pooling averages away shifts of a few hundredths of an Ångström; the embedding is concatenated with the tabular features, and training uses a Huber loss on $\log D$ plus a contrast term on the within-block pairs of each batch (Appendix A). Its block-centred prediction $s_{3\text{D}}$ enters Eq. (2) as a second shape source, $\hat{y} = a + (1 - w) s_{\text{tab}} + w s_{3\text{D}}$, with $w = 0.35$ from a nested criterion; the score becomes +0.326.

4 Results

The correlation is robust and ligand-specific. On held-out extractants, predicted and measured separation factors correlate with $r = 0.59$ (Figure 1a). A cluster bootstrap over extractants gives an interval of +0.21 to +0.38 for R_{adj}^2 . As a null test, the predictions were shuffled among extractants within each pair position. The shuffled predictions score $R_{\text{adj}}^2 = -0.05$. Removing both the position means and the extractant means (10 and 13 % of the variance) leaves the ligand-by-position interaction, which is the selectivity itself. On this term the model still scores +0.23.

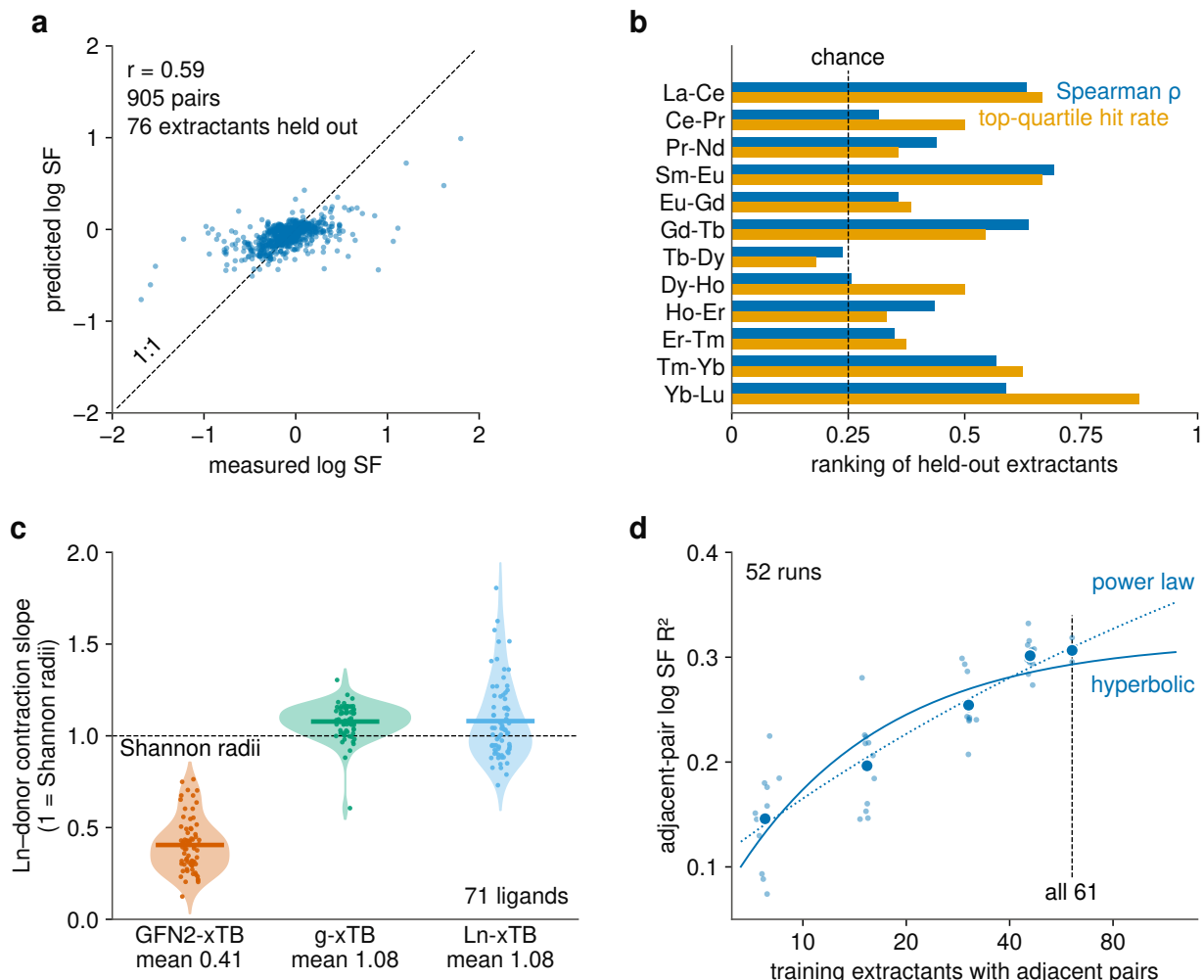


Figure 1: (a) Predicted against measured adjacent-pair logSF for the 905 pairs, extractants held out; the dashed line is the identity. (b) Per adjacent pair: Spearman ρ between predicted and measured extractant rankings, and the top-quartile hit rate against chance. (c) Slope of the Ln-donor distance against the Shannon radius (the compliance c_L) for the 71 ligands under three Hamiltonians: points are ligands, shaded shapes their kernel density, bars the mean; 1 is exact agreement with the radii. (d) Learning curve: small points are the 52 individual runs (four CV seeds, three nested draws) at their own training size, large markers the mean per fraction; solid and dotted lines the hyperbolic and power-law extrapolations; dashed line the full training set.

What the model is for. Ten percent of the pairs carry 71 % of the variance, so R_{adj}^2 is mostly a statement about the large separations. Ranking the held-out extractants for each adjacent pair, the predicted top quartile contains a measured top-quartile ligand half of the time, against a quarter by chance, and the mean Spearman correlation with the measured ranking over the twelve positions is 0.46 (Figure 1b). The same predictions detect separations above a factor of two (one pair in five) with an AUC of 0.81. Split-conformal intervals [9], calibrated leave-one-extractant-out, cover 79 % and 90 % at nominal 80 and 90. The model is a triage tool: it says which ligand classes to consider for a pair and where their large separations lie; it does not rank ligands within one class, and that boundary is part of the claim.

Table 1: Adjacent-pair log SF R^2 on the 905 pairs, extractants held out.

Model	R^2
CatBoost on log D	+0.268
3D encoder, standalone	+0.266
Level + shape, Eq. (2)	+0.318
Level + shape + 3D shape, $w = 0.35$	+0.326

5 Contribution of three-dimensional structure

The optimised complexes enter the model through the shape term only. Adding the block-centred prediction of the 3D encoder to the tabular shape with weight $w = 0.35$ raises the score from +0.318 to +0.326 (eight CatBoost seeds; 32-seed encoder ensemble), whereas the same geometries converted to 84 descriptor columns and given to a model that predicts log D directly lower it from +0.278 to -0.022 . The gain lies in the light half of the series: on the 482 La–Gd pairs the score rises from +0.356 to +0.369, while on the 347 Tb–Lu pairs it remains at +0.028. The heavy half is where every model has little skill (Spearman 0.30 against 0.58) and where the separations are smaller (sd 0.17 against 0.32).

The present geometries cannot contribute more because they carry a single number about the metal. Eight encoders with different architectures, neighbourhood cutoffs and loss functions, trained on the same complexes, make nearly identical predictions (disjoint eight-seed halves of one configuration correlate at $r = 0.990$; the eight prediction vectors have effective rank 1.05). In GFN2-xTB the lanthanide parameters are linear interpolations in Z between fitted values for Ce and Lu, so the optimised metal–donor distances vary with the metal through one scalar, equivalent to the ionic radius already among the tabular features. The contraction benchmark quantifies this (Figure 1c): for 71 ligands, all fifteen lanthanides were substituted into one relaxed complex, re-optimised with each Hamiltonian under identical settings, and the mean metal–donor distance was regressed on the Shannon radius, so that the slope c_L is 1 for a complex that contracts as the ionic radii do. GFN2-xTB gives 0.405 ± 0.145 ; g-xTB [10], with explicit f electrons, gives 1.078 ± 0.094 and is closer to 1 for 70 of the 71 ligands; Ln-xTB [11], an element-by-element re-parameterisation of GFN2-xTB, reproduces the mean slope (1.081 ± 0.214) but not the regularity of the series (a linear fit explains 43 % of its metal dependence against 82 % for g-xTB). Correcting the contraction does not improve the prediction (+0.005 with g-xTB geometries, $p = 0.68$, eight paired seeds; Appendix B), and for all three Hamiltonians the ligand-specific compliance is uncorrelated with the measured selectivity ($r = 0.11$, -0.02 and 0.17 ; 44 ligands).

What the structures lack is therefore not accuracy but ligand-specific content. Selectivity is a difference in binding free energy between two ions in the same ligand, and the quantities that could differ between ligands are absent from the present structure set by construction: the coordination number is assigned by a rule of the metal alone, so a ligand-dependent change of coordination number along the heavy half of the series cannot appear; 89 % of the complexes are the bare Ln^{3+} cation rather than the neutral extracted species; and only geometries, not the energies of accommodating each ion, are used. Structure sets that relax these choices are the natural next input for the shape channel; the compliance benchmark tests cheaply whether a candidate set carries more than one number about the metal.

How much data the problem needs. Seventy-six extractants contribute scored pairs, yet the model is not starved. Trained on random fractions of the training extractants, the score rises from +0.15 with eight extractants to +0.30 with forty-six and then levels off at +0.31 with all sixty-one (Figure 1d). The last third of the data is worth +0.005, and the two fitted extrapolations put the gain from doubling the extractant count between +0.01 and +0.04. What the descriptors lack is whatever separates ligands inside one chemical class; adding americium data runs into the same limit (Appendix D).

Limitations and conclusion. No ceiling is identified: separations reproduce to about 0.16 log units on a spread of 0.22, but 70 % of the scored pairs have no replicate. What the model does well is to say which ligands are selective and where their large separations lie, with calibrated uncertainty, for ligands it has never seen. The step that made this possible was asking for the shape of a series rather than its level.

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A Protocol

Code, out-of-fold predictions and every negative result: <https://github.com/mironovb/lanthanidestrain>.

Blocks. The ten binned condition descriptors that define a block are acid class, acid strength, nitrate activity, pH, diluent family, modifier class, temperature, contact time, phase ratio and metal concentration. Pairs with a measured logSF of exactly zero, which arise when both metals sit at a detection limit, are excluded. Promethium is absent, so Nd–Sm is not an adjacent pair.

Hyperparameters. CatBoost: 1,500 iterations, learning rate 0.04, depth 9, ℓ_2 regularisation 3, feature subsampling 0.3, quantile loss $\alpha = 0.6$, identical for the level and shape models. 3D encoder: width 96, three continuous-filter layers, 64 radial basis functions to 5 Å, cutoff 4 Å, heavy atoms only, dropout 0.15, head width 256, Adam at 2×10^{-3} with weight decay 10^{-4} , 60 epochs with early stopping on a validation fold, batches of 64 rows grouped by block so the contrast term always has pairs, pair-contrast weight 4 with adjacent pairs weighted $3\times$. Blend weight: nested selection over $\{0, 0.05, \dots, 1\}$ with equal weight per extractant.

Learning curve. Fractions 0.125, 0.25, 0.5, 0.75 and 1 of each fold’s training extractants, nested within a draw. The full-data run reproduces the stored champion exactly. Because the anchor is one number per extractant it cancels in every adjacent pair, so the level model does not affect the score and only the shape model’s training set matters.

B Controls

Interventions under the same protocol on the 905 pairs; ΔR^2 relative to the best comparable system.

Intervention	ΔR^2
Simplicial encoder instead of the distance graph	0.00 ($r = 0.96$, blend weight 0.01)
Persistence descriptors as extra shape features	−0.03 to −0.09
Metal-feature weight $\times 2$ / $\times 5$ / $\times 10$	+0.00 / −0.05 / −0.08
All-pairs difference model added to the blend	0.00
Condition-transfer features	+0.004
g-xTB geometries instead of GFN2-xTB (802 complexes)	+0.005 ($p = 0.68$)
Complexes rebuilt in correspondence	−0.013
Solvent-relaxed geometries, water / octanol	−0.008 / −0.081
GFN2 / g-xTB interaction-energy features	0.00

C Ln-xTB parameters

The Ln-xTB parameters are given in the supporting information of Zhang [11] as PDF tables only. We parsed both copies (Tables S60–S61 and S62; no disagreement), replaced the $Z = 57$ –71 blocks of the stock GFN2-xTB parameter file, and ran unmodified xtb 6.7.1. Single points on the 35 optimised structures shipped with the paper reproduce the published energies to better than $2 \times 10^{-8} E_h$ once the paper’s spin convention is used (the high-spin count of unpaired f electrons, reduced by one where its parity is inconsistent with the electron count). All 2,130 optimisations of the benchmark converged. A closed-shell control gives $c_L = 0.981 \pm 0.140$ with the same jagged series.

D Americium

SAFE holds 1,794 Am(III) rows on 110 ligands, 95% of them ligands with lanthanide data, giving 312 Am/Eu separation factors on 82 extractants under identical conditions. Zero-shot, with Am treated as a lanthanide at its radius, the Am/Eu separation is not predicted ($R^2 = -0.14$). Joint training with the Am rows reaches +0.47 on held-out extractants, but a control that assigns each held-out extractant the mean of the other extractants in its ligand family scores +0.47 as well, and within a family the model has no skill (+0.02); the lanthanide score is unchanged ($+0.002 \pm 0.020$ over paired seeds). The model learns which ligand classes discriminate Am from Eu, nothing finer.